

Advanced Encryption Standard (AES)

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Presentation Overview

Defining AES – What is AES, and why was it created

Methodology – How does AES encryption and decryption work

Real World Use Cases – Where is AES used, and what roles does it play in your daily lives

Future of AES – The current outlook of AES and its security

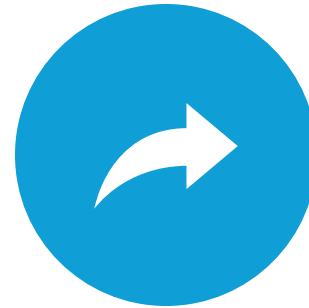
Background



AES IS A SYMMETRIC CIPHER
STANDARD THAT WAS ADOPTED BY
THE U.S GOVERNMENT IN 2002 AS
A REPLACEMENT FOR 3-DES



AES CAN HANDLE KEYS OF 128,
192, 256 BITS TO ENCRYPT AND
DECRYPT DATA IN BLOCKS OF 128
BITS



REPLACEMENT FOR DES / 3-DES

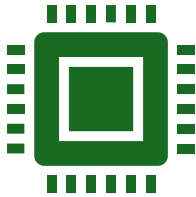


USED IN BOTH GOVERNMENT AND
CONSUMER SECTORS FOR
ENCRYPTION

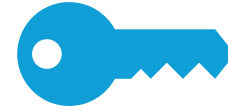
Introduction to DES



The Data Encryption Standard (DES) was AES' predecessor



Adopted in 1977 by the National Bureau of Standards (later known as NIST) for implementation in electronic hardware devices and cryptographic protection of computer data



64 bit key – 56 random bits for algorithm use, 8 bits for error detection. Worry about this key size was immediate, with the 1977 NIST Workshop on Cryptography concluding a 48-bit key would be unacceptable, and that though 56-bit is acceptable a 64-bit key would be preferable

Methodology of DES

Steps

IP

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

Initial permutation

IP⁻¹

40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25

Inverse initial permutation

1. Take your 64-bit input and reorder it according to IP
2. Split the reordered input in half (L, R)
3. Apply no changes to R
4. Take L and xor it with the output of $f(R, K)$ where f is a round function that produces a block of 32 bits from the 32-bit R and the 48-bit K (portion of your key)
5. The next step considers the previous' L as R and R as L - repeat for 16 total rounds
6. Take that output and reorder according to IP⁻¹ to get the final ciphertext

This can be deciphered by applying the exact same process in reverse to the ciphertext, ensuring the same K is used each step

FIPS PUB 46

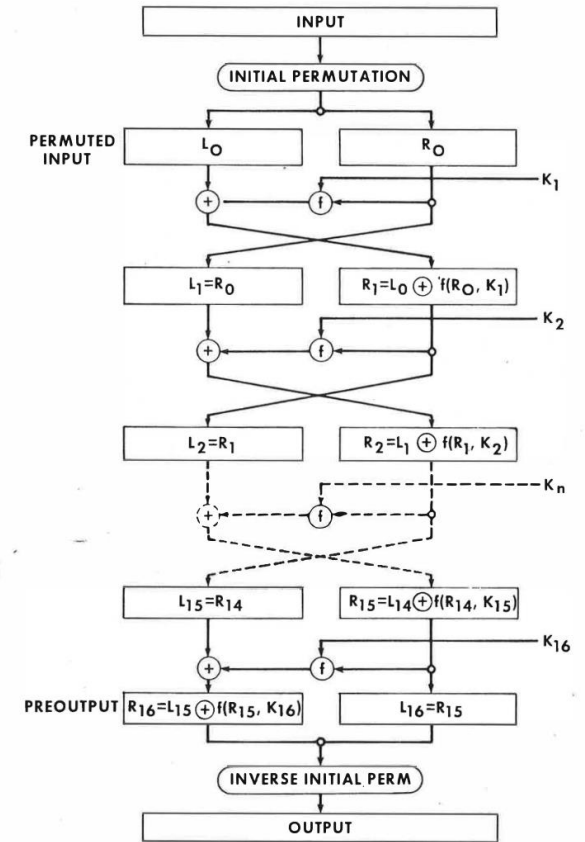


FIGURE 1. Enciphering computation.

A diagram explaining DES from the original NBS standard

Downfall of DES

DES Cracks by Year

Year	Method	Time to Crack	Difference from prior year
1997	Brute-force using a network of computers	96 days	n/a
1998	"Deep Crack" device to recover keys	56 hours	41.14x faster
1999	"Deep Crack" + computer network	22.25 hours	1997 – 104.01x faster 1998 – 2.49x faster

- If done properly there are 70,000,000,000,000,000 (seventy quadrillion) possible keys of 56-bits
 - Even in the initial NBS document they state, "The feasibility of computing the correct key may change with advances in technology."
- The worry of technological advances was proven true when DES was cracked

Downfall of DES

- In 1999 following the key being cracked so quickly NIST encouraged transition from DES to 3-DES, though DES continued to be approved until May 2005
 - 3-DES is a process using 2 or 3 independently generated keys to encrypt by – encrypting -> decrypting -> encrypting
 - Computationally inefficient, still based on outdated method
- NIST had begun putting out calls for a new algorithm in 1997
 - The Rijndael algorithm was eventually selected in October 2000
 - In December 2001 NIST announced the approval of AES as "Federal Information Processing Standard" - finishing the process of making the standard

AES Methodology

- Symmetric block cipher
- Operates on a fixed block size of 128 bits (16 bytes)
- Supports three key lengths:
 - 128-bit key -> 10 rounds
 - 192-bit key -> 12 rounds
 - 256-bit key -> 14 rounds
- Each round applies mathematical transformations to strengthen encryption by obscuring data patterns and spreading changes across the entire block.



AES Internal Structure

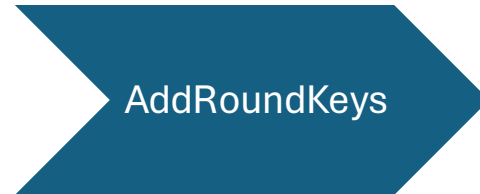
State Matrix Representation

- AES stores plaintext as a 4x4 byte matrix, called the State Matrix
- The matrix undergoes multiple transformations during encryption
- Encryption begins with an initial AddRoundKey operation
- Followed by several other transformation rounds

B0	B4	B8	B12
B1	B5	B9	B13
B2	B6	B10	B14
B3	B7	B11	B15

Initial Step:

- Plaintext is XOR'd with the initial round key



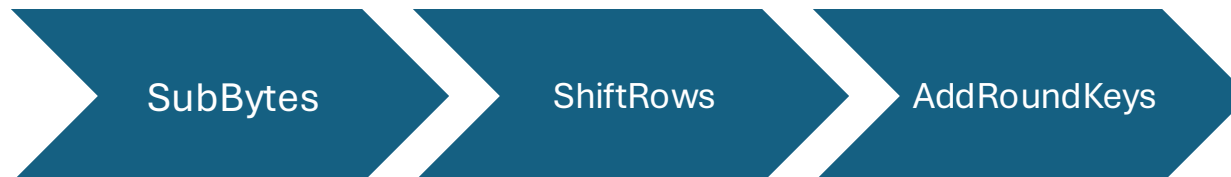
Main Rounds:

- Each round performs 4 steps:



Final Rounds:

- Same operations except MixColumns is omitted



AES Type	Rounds
AES-128	10
AES-192	12
AES-256	14

- The number of rounds performed depends on the type of AES used

AES Key Expansion

- AES expands the original encryption key into multiple Round Keys
- This process is called Key Expansion or Key Scheduling

Example:

- 128-bit key
 - Expanded into 44 words (32-bit each)
 - Produces 11 round keys

Purpose

- Provides a unique key for every encryption round
- Strengthens resistance against attacks

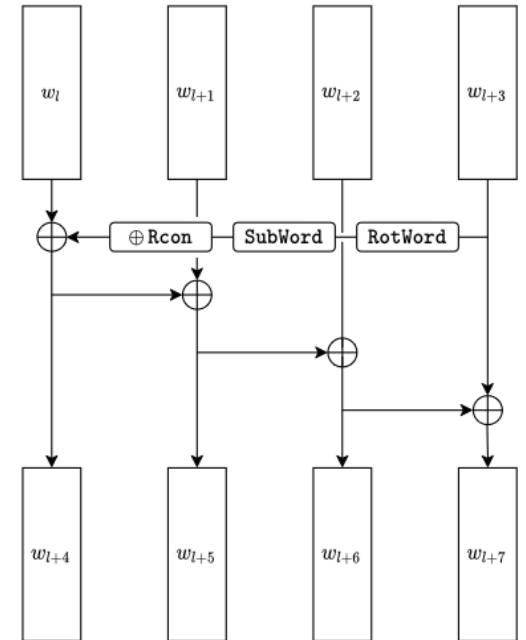
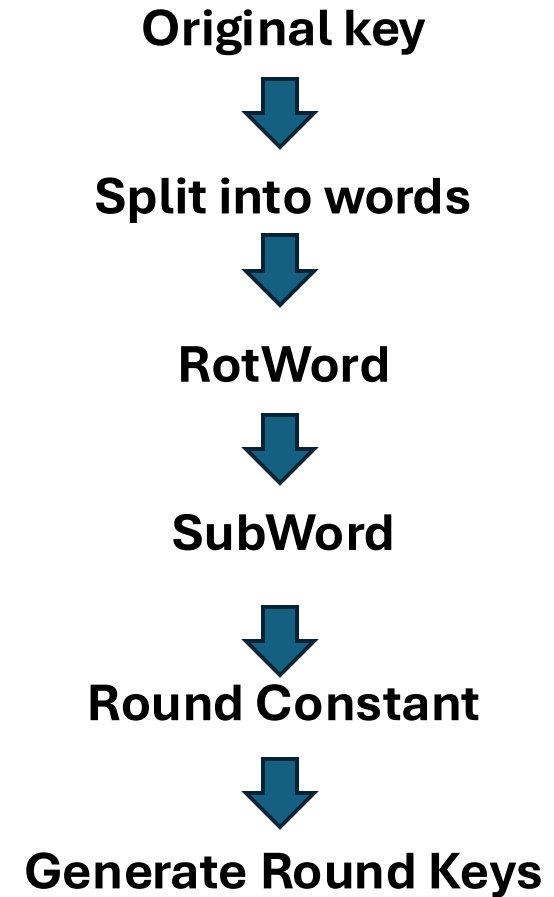


Figure 6. KEYEXPANSION() of AES-128 to generate the words $w[i]$ for $4 \leq i < 44$, where l ranges over the multiples of 4 between 0 and 36

Illustration from NIST FIPS 197

Key Expansion

- Start with the original cipher key
- Split key into 32-bit words
- Apply transformations:
 - RotWord: rotates bytes
 - SubWord: substitutes bytes using S-box
 - Rcon: adds round constants
- Generate new round keys iteratively
- Does not rely on a single repeated key



SubBytes Operation

- Nonlinear byte substitution
- Each byte in the State matrix is replaced using Substitution box
- The S-box is a predefined lookup table

Purpose:

- Introduces nonlinearity
- Protects against statistical and algebraic attacks
- Increases encryption complexity

Result:

Input bytes are transformed into unrelated output values

S(b0)	S(b4)	S(b8)	S(b12)
S(b1)	S(b5)	S(b9)	S(b13)
S(b2)	S(b6)	S(b10)	S(b14)
S(b3)	S(b7)	S(b11)	S(b15)

Shift Rows Operation

Rows of the State Matrix are cyclically shifted left

Row 0: No shift

Row 1: 1 byte

Row 2: 2 bytes

Row 3: 3 bytes

Purpose:

- Rearranges byte positions
- Improves diffusion
- Prevents columns from remaining unchanged

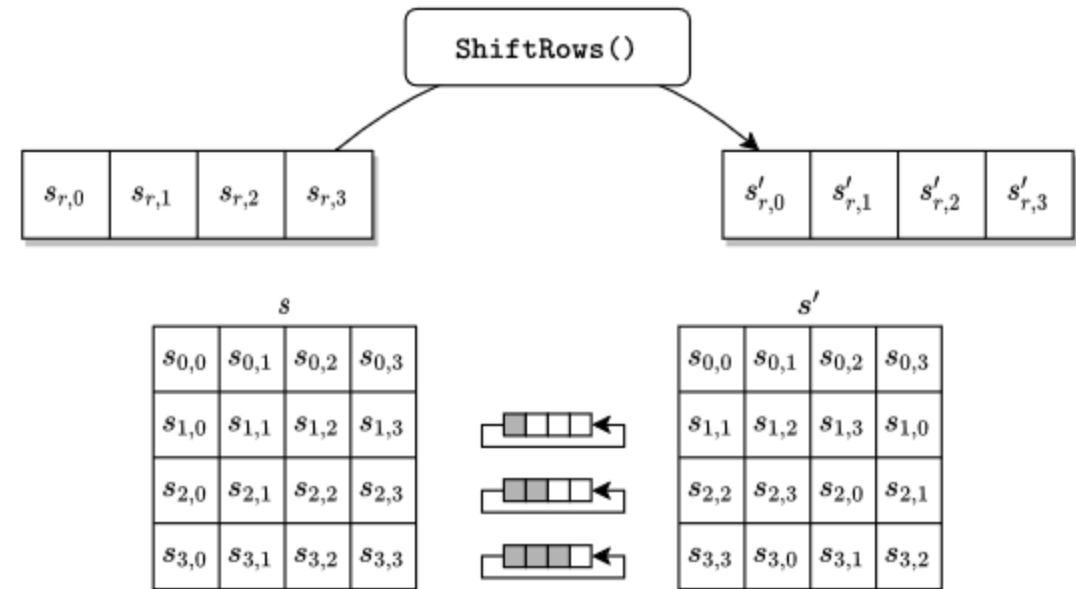


Figure 3. Illustration of SHIFTRows()

Illustration from NIST FIPS 197

MixColumns Operation

- Each column is transformed using Galois Field arithmetic ($GF(2^8)$)
- Matrix multiplication mixes bytes within each column

Purpose:

- Spreads influence of each byte across the column
- Enhances diffusion
- Makes patterns harder to detect

This step is not performed in the final AES round

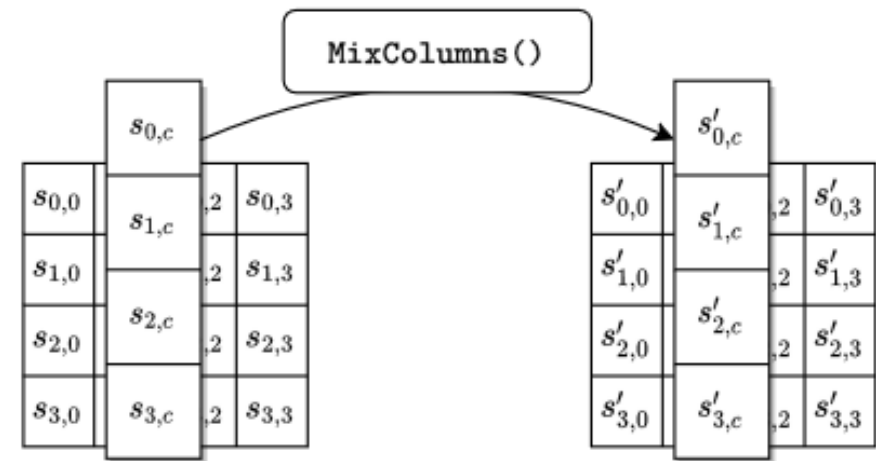


Figure 4. Illustration of MIXCOLUMNS()

Illustration from NIST FIPS 197

AddRoundKey Operation

The State Matrix is combined with a Round Key using:

- Bitwise XOR operation

Purpose:

- Integrates encryption key material into every round
- Ensures encryption depends directly on the secret key

Formula:

State = State XOR RoundKey

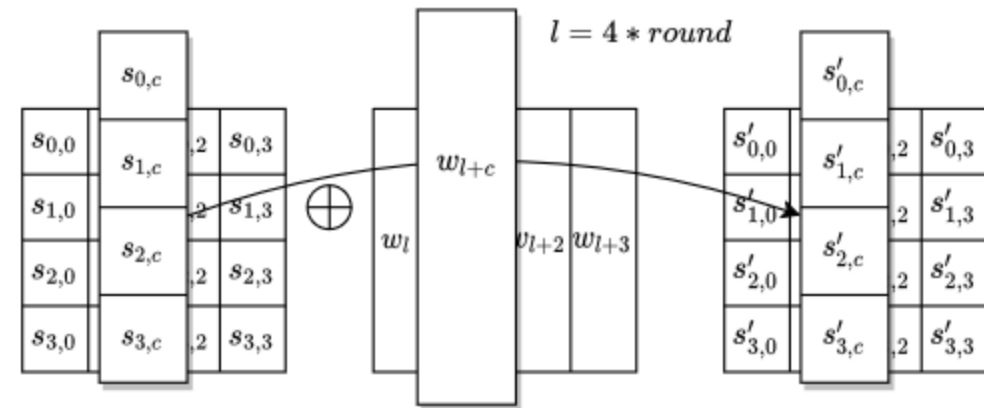


Figure 5. Illustration of ADDROUNDKEY()

Illustration from NIST FIPS 197

AES DEMO

AES Decryption Process

AES decryption applies inverse operations:

- Inverse SubBytes
- Inverse ShiftRows
- Inverse MixColumns
- AddRoundKey (Round keys are used in reverse order)

Result:

- Original plaintext is reconstructed from ciphertext

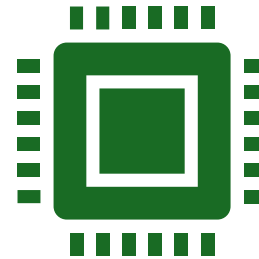
DES vs AES



Most important - 56-bit key (DES) vs 128-bit(+) key (AES)

Allows AES to be faster than DES since it can process info faster, making AES more efficient and scalable

Greater security – 2^{56} potential keys (DES) vs 2^{128} or 2^{192} or 2^{256} AES(128,192,256)



AES is computationally heavier due to requiring more complex rounds of transformation and the larger key size

Allows DES to run in some situations AES may not be able to (e.g., embedded or legacy systems)

Security drawbacks and the average device having enough computation power make this point largely moot

Strengths of AES

- Resilient against brute force – Even if the "DES Cracker" device could crack a DES key in one second it would still take approximately 149 trillion years to crack AES-128
- Open nature allows cybersecurity experts to be on a constant look out for potential vulnerabilities

Weaknesses of AES

- In 2009 a cryptanalysis discovered attackers could try to crack a system by seeing how it operates using different keys
 - Experts have concluded this can only happen in AES systems that are not properly configured
- Vulnerable to side-channel attacks where attackers try to pick up information leaking from the system to see how it works
 - Can only happen in non-secure, improperly configured systems
- Dependent on proper configuration and security of the key – vulnerable to human error

Key Management

- AES security depends on protecting the encryption key
- Weak or exposed keys compromise otherwise secure systems
- Secure key exchange and storage are critical
- Solutions:
 - **Public Key Infrastructure (PKI)** - Method of securely exchanging keys where the key is encrypted with a public encryption key, and decrypted with a private decryption key
 - **Hardware Security Modules (HSMs)** - A physical computing device that protects and manages a private key. Used for secure key storage.
 - **Key rotation policies** – Rotating used keys to limit the chances for attackers to gain information on the system of encryption

Mode of Operation

- AES is a block cipher and requires a mode of operation to securely handle data larger than 128 bits
 - These modes apply to any symmetric key block cipher algorithm
- Some modes are insecure if used improperly

Modes:

- ECB – Simple and encrypts each block identically.
- CBC – A forward cipher in which the output of the prior block is XOR'd with the current block's plaintext before the cipher is applied
- GCM - uses universal hashing over a binary Galois field to provide authenticated encryption

Mode of Operation

Which mode is recommended?

Mode	Status	Notes
ECB	Insecure	Reveals patterns
CBC	Not the best choice	Confidentiality not integrity
GCM	The Modern Standard	Encryption + authentication

Performance vs Security

- Modern networks require fast encryption at large scale
- AES benefits from hardware acceleration
 - AES-NI – Hardware implementation on the CPU, "used to accelerate the performance of an implementation of AES by 3 to 10x over a completely software implementation" (Intel)
- Stronger security can increase computational overhead
- Systems must balance:
 - Speed
 - Scalability
 - Security requirements

The Future of AES

What does Quantum Computing mean for the future of AES?

Current state:

- No practical attacks currently exist against properly implemented AES
- AES-128 and AES-256 remain industry standards
- Used globally in:
 - HTTPS
 - VPNs
 - Banking systems

The Future of AES

Quantum Computing Threats

- Quantum computing could reduce AES security using Grover's Algorithm
 - Grover's algorithm can search large datasets faster than classic methods – meaning a brute force attack on a 128-bit key could theoretically require 2^{64} operations instead of 2^{128}
- NIST has said that AES-128 remains secure in a post-quantum context
 - Position reinforced by other cryptographic researchers
- Quantum is more likely to threaten public key exchange methods which could force AES key exchanges to be managed differently or introduce vulnerabilities if threat isn't responded to promptly

The Future of AES

Future Trends:

- Increased dependency on AES-256 over shorter AES-128
- Development of post quantum cryptography
- Combining AES with quantum resistant algorithms??
- Continued hardware optimization for encryption performance

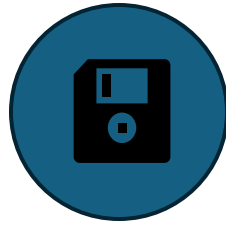
Real World Uses of AES



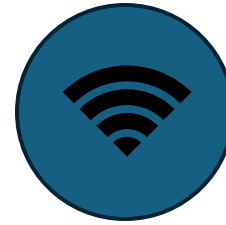
Archive and
compression



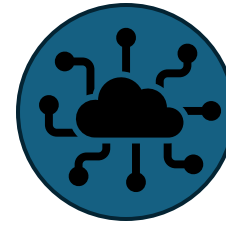
File encryption
systems



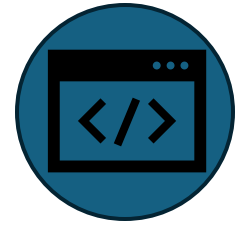
Disk and partition
encryption



Local area network
security
(WPA2/WPA3)



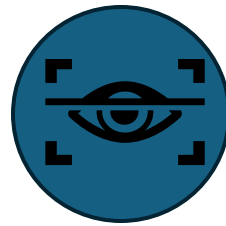
Internet Protocol
Security (IPsec)



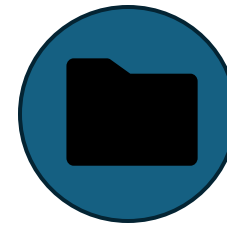
Linux kernel's
Application
Programming Interface
(API)



Message encryption in apps
like Signal/Facebook
Messenger(Optional)



Virtual Private
Networks (VPNs)



Secure file transfer
services like Kiteworks

Sources

- Source for DES information <https://csrc.nist.gov/files/pubs/fips/46/final/docs/nbs.fips.46.pdf>
- Source for DES vs AES slide: <https://ijrpr.com/uploads/V6ISSUE5/IJRPR45409.pdf>
- Source for AES strengths: <https://csrc.nist.gov/encryption/aes/aesfact.html>
- Source for signal: <https://capilano-ics.com/2025/03/29/understanding-signals-encryption-a-technical-breakdown/>
- Source for others uses: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9682931/> (under widespread AES use)
- Source for modes: <https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-38a.pdf>
- Source for information about quantum computing threats: <https://thequantuminsider.com/2026/04/21/cryptologist-finds-aes-128-likely-safe-from-quantum-attack/>
- Intel about hardware components: <https://www.intel.com/content/www/us/en/developer/articles/technical/advanced-encryption-standard-instructions-aes-ni.html>
- GCM <https://csrc.nist.gov/groups/ST/toolkit/BCM/documents/proposedmodes/gcm/gcm-revised-spec.pdf>
- More info about GCM <https://csrc.nist.gov/groups/ST/toolkit/BCM/documents/proposedmodes/gcm/gcm-spec.pdf>